SYSTEM DESIGN

CAP Theorem Visual Guide

Consistency, Availability, Partition tolerance: why distributed systems must pick two.

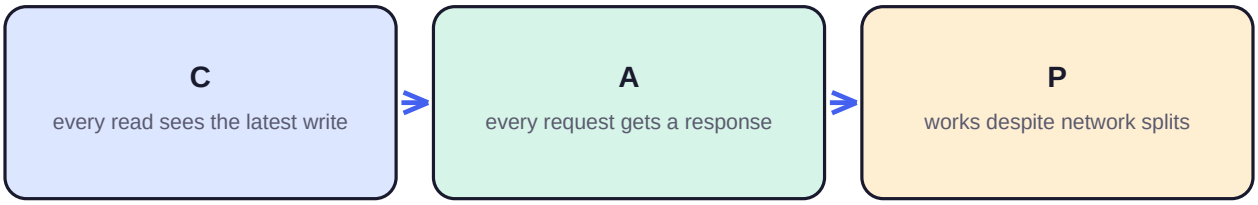
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

The Three Properties

Every distributed data store promises some of each.



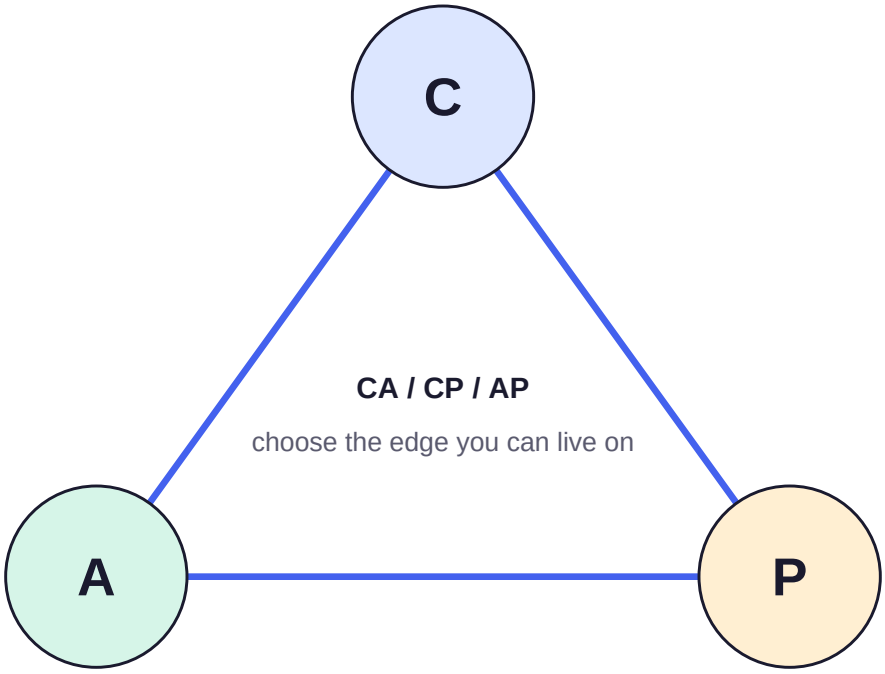
- Consistency (linearizability): all nodes agree, always -- like one machine.
- Availability: the system answers every request, even if some nodes are sick.
- Partition tolerance: a network split cannot stop the system from operating.

These are guarantees about BEHAVIOR during failures, not features you toggle.

STEP 2/6

Pick Two

During a partition, you can only guarantee two of the three.

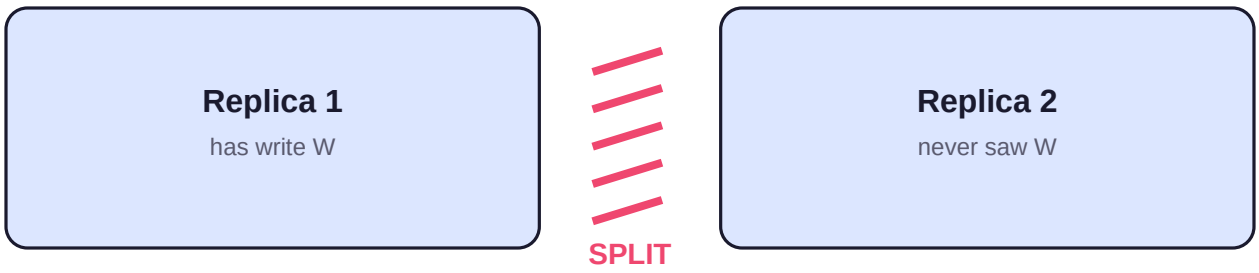


CA (no P) exists only if you deny partitions happen -- true of a single node, not a distributed system.

STEP 3/6

Partitions Are Not Optional

Networks WILL split. The question is what you sacrifice when they do.



- Choice 1 (CP): Replica 2 REFUSES reads until it can confirm W -- availability lost.
- Choice 2 (AP): Replica 2 answers from stale data -- consistency lost.
- There is no third choice. The partition forces your hand.

STEP 4/6

CP Systems: Consistency First

Block or fail rather than serve a stale answer.

etcd / ZooKeeper

consensus quorum; minority partition stops writes

HBase

single region-server per range; split = unavailable

MongoDB (majority)

writes need majority ack; stale reads rejected

Use CP for: leader election, config, financial ledgers -- anywhere wrong data beats no data.

STEP 5/6

AP Systems: Availability First

Answer always; reconcile conflicts later.

Cassandra

tunable consistency; hinted handoff replays writes

DynamoDB

leaderless replication, vector clocks, sloppy quorum

CouchDB

multi-master; conflicts kept as document revisions

Use AP for: shopping carts, feeds, session state -- anywhere downtime costs more than staleness.

STEP 6/6

PACELC: The Missing Half

CAP only covers partitions. PACELC asks: what about normal operation?

P A C E L C

Partition? -> A or C. Else -> Latency or Consistency.

- Even with NO partition, synchronous replication costs latency on every write.
- Systems like Dynamo choose Latency: async replication, stale reads possible even when healthy.
- Systems like Spanner choose Consistency: TrueTime waits out clock uncertainty (small latency tax).
- So the real trade-off is four-way: partition behavior AND steady-state latency vs consistency.

STEP ✓I✓

Key Takeaways

CAP Theorem — Remember

- C = every read sees the latest write; A = every request answered; P = survives network splits.
- During a partition you keep exactly two: CP blocks rather than lie, AP answers possibly-stale.
- Partitions are inevitable, so 'CA' only describes a single machine -- not a distributed system.
- PACELC completes the picture: with no partition, you trade Latency against Consistency.
- Choose by workload: ledgers and locks want CP; carts and feeds want AP.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev